

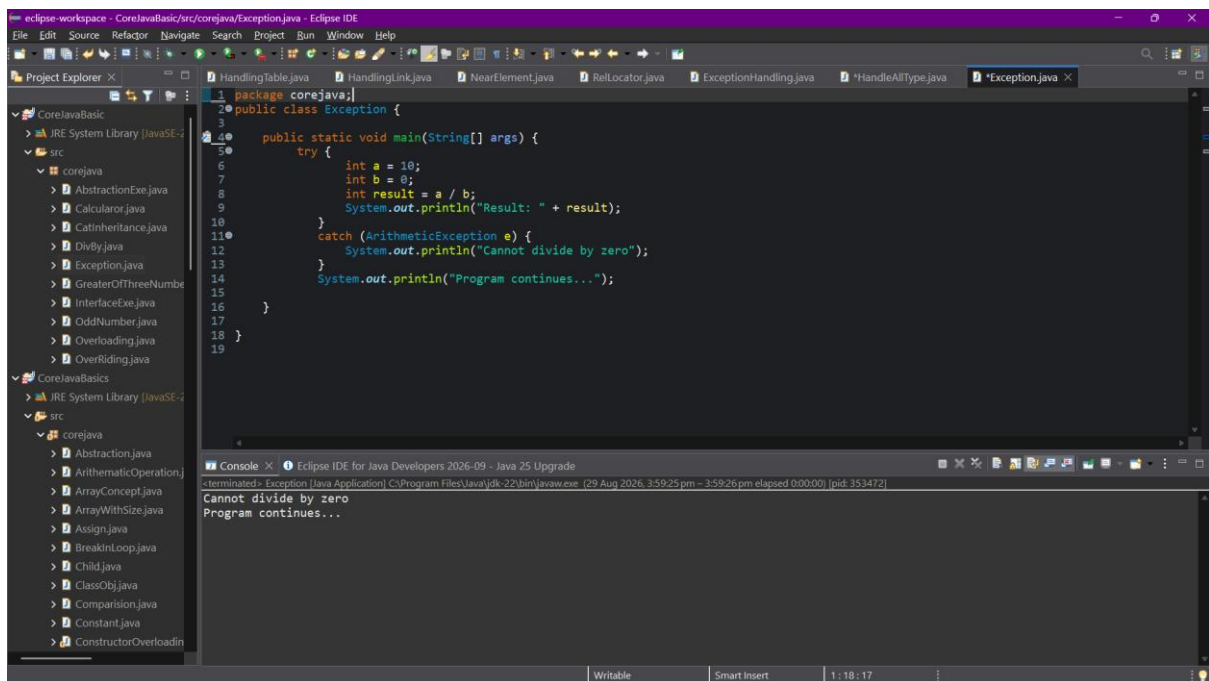
Automation testing

Day-07 Assignment

1.What is Exception? Write a program for exception handling:

Exception :

An **exception** is an unexpected event or error that occurs during the execution of a program and **disturbs the normal flow of the program.**



The screenshot shows the Eclipse IDE interface. The Project Explorer on the left displays a project named 'CoreJavaBasic' with a package 'corejava' containing several files, including 'Exception.java'. The main editor window shows the code for 'Exception.java'.

```
1 package corejava;
2 public class Exception {
3
4     public static void main(String[] args) {
5         try {
6             int a = 10;
7             int b = 0;
8             int result = a / b;
9             System.out.println("Result: " + result);
10        }
11        catch (ArithmeticException e) {
12            System.out.println("Cannot divide by zero");
13        }
14        System.out.println("Program continues...");
15    }
16 }
17
18 }
19 }
```

The Console window at the bottom shows the output of the program:

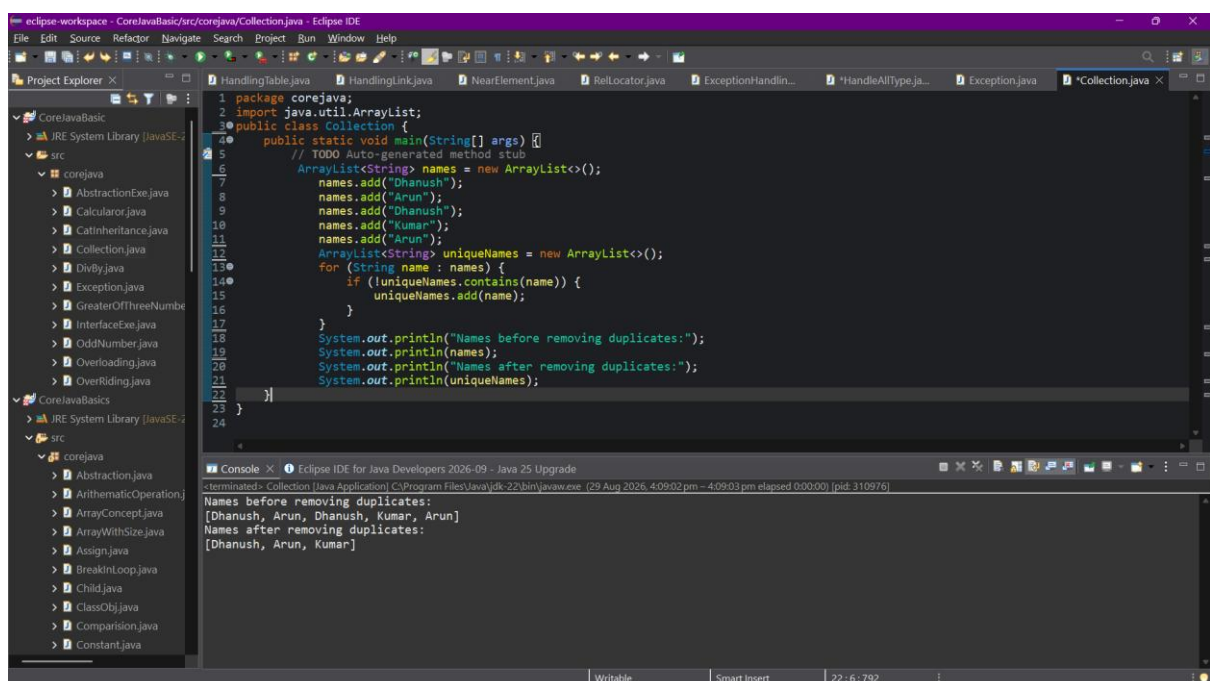
```
Terminated - Exception [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (29 Aug 2026, 3:59:25 pm - 3:59:26 pm elapsed 0:00:00) [pid: 353472]
Cannot divide by zero
Program continues...
```

2.What is Collections? Write a program to remove duplicate names using Collections

Collections in Java are used to store and manage a group of objects.

Java provides the **Collections Framework**, which contains classes and interfaces such as:

- ArrayList – stores elements in a list.
- HashSet – stores unique elements and removes duplicates.
- HashMap – stores data in key-value pairs.
- LinkedList – stores elements as a linked list.



The screenshot shows the Eclipse IDE with a project named 'CoreJavaBasic'. The 'src' folder contains several Java files, including 'Collection.java'. The 'Collection.java' file is open, showing the following code:

```
1 package corejava;
2 import java.util.ArrayList;
3 public class Collection {
4     public static void main(String[] args) {
5         // TODO Auto-generated method stub
6         ArrayList<String> names = new ArrayList<>();
7         names.add("Dhanush");
8         names.add("Arun");
9         names.add("Dhanush");
10        names.add("Kumar");
11        names.add("Arun");
12        ArrayList<String> uniqueNames = new ArrayList<>();
13        for (String name : names) {
14            if (!uniqueNames.contains(name)) {
15                uniqueNames.add(name);
16            }
17        }
18        System.out.println("Names before removing duplicates:");
19        System.out.println(names);
20        System.out.println("Names after removing duplicates:");
21        System.out.println(uniqueNames);
22    }
23 }
24 }
```

The console output shows the following:

```
<terminated> Collection [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (29 Aug 2026, 4:09:02 pm - 4:09:03 pm elapsed 0:00:00) [pid: 310976]
Names before removing duplicates:
[Dhanush, Arun, Dhanush, Kumar, Arun]
Names after removing duplicates:
[Dhanush, Arun, Kumar]
```